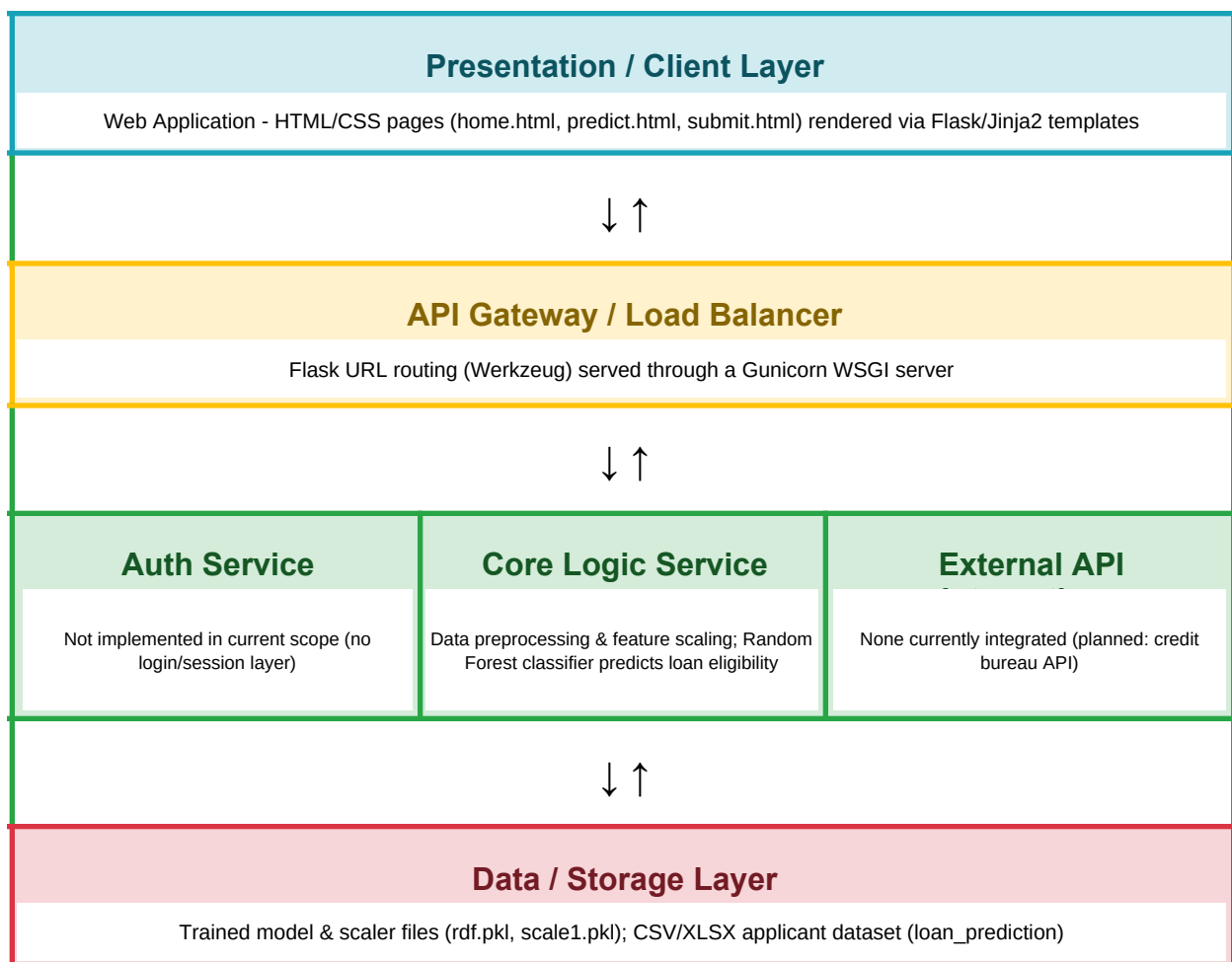


Solution Architecture

Date	15 March 2026
Team ID	
Project Name	SmartLender - Applicant Credibility Prediction for Loan Approval
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>[Presentation Layer]</i>	Renders the user interface for entering applicant details and displaying the loan prediction result.	HTML, CSS, Flask (Jinja2 templates)
<i>[API Gateway]</i>	Routes incoming HTTP requests between the client and the core prediction logic; serves the application.	Flask routing, Gunicorn (WSGI server)
<i>[Microservices]</i>	Preprocesses applicant input, scales features, and runs the trained ML model to predict loan approval.	Python, scikit-learn (RandomForestClassifier), NumPy, Pandas, pickle
<i>[Database]</i>	Stores the historical applicant data used to train the model, and the serialized model/scaler files.	CSV/XLSX dataset (loan_prediction), local .pkl file storage